

Branch-Dependent Proper Time

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Abstract

We propose tying the fundamental scale of gravitationally induced space-time blurring to *proper time* rather than to spatial length, and postulate that, under an ensemble reading of the quantum state in which each branch carries its own world line, proper-time fluctuations are statistically independent across branches. Interference is sensitive to the *difference* of branch phase contributions; with independent clocks this difference accumulates at the frequency $\omega_{\text{eff}} = E_{\text{br}}/\hbar$. For a composite system we define this branch-sensitive energy operationally: the rest energy of each particle species is weighted by the trace distance between the one-particle reduced density matrices of the two branches; a fully distinguishable which-path carrier gives $E_{\text{br}} = mc^2$, while strictly internal branches with a common spatial-motional state give $E_{\text{br}} = \Delta E$. This yields $V/V_0 = \exp(-\chi)$ with $\chi = (\omega_{\text{eff}})^2 (\xi t_P^2 t)^{2/3}$ for a branch pair with $E_a \approx E_b$ (for internal branches with $E_a = 0$ an additional factor $1/2$ appears); for a centre-of-mass interferometer with fully distinguishable packets $\chi = (mc^2/\hbar)^2 (\xi t_P^2 t)^{2/3}$, where t is the superposition time and $\xi = \mathcal{O}(1)$ is a numerical coefficient. The scaling form $\chi \propto m^2 t^{2/3}$ follows from the postulates; the overall coefficient carries an $\mathcal{O}(1)$ uncertainty. The cubic scaling is borrowed from the Károlyházy relation; the decoherence mechanism differs from Diósi–Penrose collapse. Confronted with sodium-nanoparticle interferometry of the Vienna group [1], this gives an upper bound $\xi \lesssim 1.1$. A decisive test is near: at $m \sim 1$ MDa and $v \approx 25$ m/s the model predicts $\chi \gg 1$ and practically complete loss of interference, whereas the standard quantum model of the setup at the same parameters expects measurable fringes with ordinary contrast.

Keywords: proper time, branch clock independence, Károlyházy uncertainty, decoherence, matter-wave interferometry, Margolus–Levitin theorem

1 Introduction

The status of the wave function—an ontic object or a computational bookkeeping of an ensemble of possibilities—remains unsettled. We adopt the second, ensemble reading [2, 3]: $|\psi\rangle = \sum_i c_i |i\rangle$ catalogues distinct *variants* of the system, and the squared amplitudes are the relative weights of those variants. This is a deliberately minimal ontology; the substantive postulates on independent branch clocks (§ 2) are added separately, once gravity enters the picture.

In the relational view of time, a clock is a physical subsystem and “time” is a correlation within the total state [4]. If each branch is a genuine variant of the system, then it carries its own world line and hence its own proper time. Penrose reaches a structurally identical statement from the opposite direction: each component of the superposition sources its own space-time, and “since there is no canonical way of relating the time parameters in the two space-times, a fundamental uncertainty in

the time-translation operator arises” [5]. Consistently with this, proper time accumulates along a world line and is not fixed by a single common scale for distinct four-dimensional paths: in an interferometer with different trajectories one cannot, without an additional postulate, impose a single common proper-time parameter on all branches. While the system is in a superposition of paths, interference requires a coherent relative phase between them; branch-independent proper-time fluctuations (postulates P2–P3 below) introduce a random shift of this phase and bound the visibility from above, $V/V_0 \leq e^{-\chi}$ (see (6))—this is a prediction of an interference limit under path-dependent proper time, not a denial of amplitude superposition. After a which-path registration, a single scenario and a single realized world line are fixed; the $|c_i|^2$ remain the weights of the outcomes. The mechanism below is pure dephasing between branches: populations are not redistributed, and no which-path information is recorded.

2 Postulates

We introduce the key notation:

- t — the time the system spends in superposition (a proper-time interval along the world line);
- δt — the scale of proper-time uncertainty over the interval t ; δt_a — the realized shift on branch a (P3), $\text{Var}(\delta t_a) = \delta t^2$;
- $t_P = \sqrt{\hbar G/c^5}$ — the Planck time;
- E_{br} — the energy of the *split* subsystem in the interfering branch pair;
- $\omega_{\text{eff}} = E_{\text{br}}/\hbar$ — the corresponding effective phase-noise frequency;
- m — the carrier mass in the which-path case (then $E_{\text{br}} = mc^2$, $\omega_{\text{eff}} = \omega_C = mc^2/\hbar$);
- $\xi = \mathcal{O}(1)$ — a dimensionless numerical coefficient.

2.1 (P1) Branch structure.

A branch is a state component differing in a *physical degree of freedom* that participates in interference: a spatial-motional state (which-path or current), or an internal eigenstate of H_{int} (spin, a hyperfine or optical level, a qubit). One connected and indistinguishable wave packet carries one clock; branching occurs when evolution splits the state into distinguishable components, and

ceases upon their recombination or which-path registration.

To prevent the choice of the “split subsystem” from depending on an arbitrary grouping of a composite system, we fix an operational rule. For particles of species s , let $\gamma_{a,s}^{(1)}$ and $\gamma_{b,s}^{(1)}$ be the one-particle reduced density matrices of the branches, normalized by $\text{Tr} \gamma_{a,s}^{(1)} = \text{Tr} \gamma_{b,s}^{(1)} = N_s$. The number of one-particle states distinguishable between the branches and the corresponding branch-sensitive energy are

$$\mathcal{D}_s(a, b) = \frac{1}{2} \left\| \gamma_{a,s}^{(1)} - \gamma_{b,s}^{(1)} \right\|_1, \quad E_{\text{br}}^{\text{mot}} = c^2 \sum_s m_s \mathcal{D}_s(a, b). \quad (1)$$

Here $0 \leq \mathcal{D}_s \leq N_s$; the definition is basis-invariant and applies to identical particles. For fully distinguishable spatial packets of every constituent, $\mathcal{D}_s = N_s$, and hence $E_{\text{br}}^{\text{mot}} = Mc^2$. States common to both branches contribute zero. If only an internal level differs while the spatial-motional carrier $|C\rangle$ is common, we use the transition energy $E_{\text{br}}^{\text{int}} = |\Delta E|$, not the rest energy of the common carrier. If both types of difference are present, their branch-sensitive contributions are added.

Equation (1) sharpens P1 rather than following from P2–P3; the trace-distance choice [6] is a testable part of the model, and a different distinguishability measure cannot be hidden in $\xi = \mathcal{O}(1)$.

2.2 (P2) Primacy and magnitude of the proper-time fluctuation.

The accumulated uncertainty of the proper-time interval t measured along a world line is

$$\boxed{\delta t^3 = \xi t_P^2 t}, \quad (2)$$

with $\xi = \mathcal{O}(1)$. Equation (2) is the temporal form of the Károlyházy uncertainty relation [7]; the same cubic scaling is consistent with the clock arguments of Salecker–Wigner [8], Ng–van Dam [9, 10], and the holographic bound on space-time uncertainty [11]. We do not fix the numerical coefficient ξ in the postulate: in these works it is likewise never pinned down more precisely than to $\mathcal{O}(1)$.

We stress that gravity is essential: without the “not a black hole” constraint the effect vanishes

($\delta t \rightarrow 0$ as $m \rightarrow \infty$). The rest mass is equally essential: the uncertainty δt is defined along a world line with nonzero proper time; without rest mass there is no proper time. For a photon $m = 0$, $d\tau = 0$, and the entire mechanism does not apply to a light-like carrier. The exponent $1/3$ and the Planck scale t_P^2 are gravitational in origin; the postulate is formulated along a world line, without explicitly invoking spatial length.

2.3 (P3) Branch clock independence.

For two branches a, b (in the sense of P1) over the superposition interval t (the same as in (2)), the realized shifts $\delta t_a, \delta t_b$ are independent, centered Gaussian random variables of accumulated proper time with $\text{Var}(\delta t_a) = \text{Var}(\delta t_b) = \delta t^2$ and $\text{Cov}(\delta t_a, \delta t_b) = 0$. This is a substantive postulate, motivated by the ensemble ontology (§ 1): distinct variants do not share a common clock; there is no strict derivation from it, and what remains irreducible is the premise that the holographic information budget is allocated *per branch* rather than as a single shared register. Within one branch, all degrees of freedom *not* participating in the split share that branch's clock.

Crucially, δt_a is not a fluctuation of an external metric field and not environmental noise common to nearby trajectories. It is an *ontic* random shift of proper time; its scale over the interval t is set by δt from (2)—the holographic scale of such uncertainty [11]. An environmental source—for example, a difference of gravitational potentials on the interferometer arms—would give a *correlated* shift δt_a , the closer to common the nearer the branches, and it would cancel in the relative phase; but that is a systematic shift, not noise. The holographic uncertainty attaches to the split subsystem of a branch, not to spatial points, so spatial proximity of branches does not by itself correlate their clocks. Attachment to discrete branches rather than positions is essential: noise independent for arbitrarily close *positions* would mean a random phase field with unbounded gradient—hence a force and momentum diffusion; noise independent for discrete *branches* is pure dephasing without force (§ 4.1).

The law $\text{Var}[\delta t(t)] \propto t^{2/3}$ fixes only the single-interval variance and does *not* by itself determine a multi-time covariance, noise spectrum, or spin-echo response. If Gaussianity of the full

process, self-similarity, and stationary increments are additionally imposed, the natural completion is fractional Brownian motion with Hurst exponent $H = 1/3$ [12]. In that completion, disjoint increments are anticorrelated, the low-frequency spectrum of fractional Gaussian noise behaves as $S(f) \propto |f|^{1/3}$, and echo requires an explicit calculation with the corresponding filter function rather than an automatic cancellation of a quasistatic phase. These spectral properties follow from an additional temporal postulate, not from P2–P3 as presently stated.

3 Decoherence functional

For a composite system we use the decomposition $H = mc^2 + P^2/(2m) + H_{\text{int}}$. For which-path ($E_{\text{br}} = mc^2$ by (P1)) the phase contribution of the split subsystem gives the relative phase

$$\Delta\varphi = \frac{E_a}{\hbar} \delta t_a - \frac{E_b}{\hbar} \delta t_b \approx \frac{mc^2}{\hbar} (\delta t_a - \delta t_b), \quad (3)$$

since $E_a \approx E_b \approx mc^2$ and interference involves the *difference* of contributions, not the absolute phase of each branch.

The general formula follows from (3) and postulates (P2)–(P3). By (P3), with $\text{Var}(\delta t_a) = \text{Var}(\delta t_b) = \delta t^2$:

$$\text{Var}(\Delta\varphi) = \frac{E_a^2 + E_b^2}{\hbar^2} \delta t^2. \quad (4)$$

If the centre of mass branches (which-path), (4) gives

$$\text{Var}(\Delta\varphi) = 2\omega_{\text{eff}}^2 \delta t^2, \quad \omega_{\text{eff}} = \frac{mc^2}{\hbar}. \quad (5)$$

With *shared* clocks the global phase $mc^2\tau/\hbar$ would cancel in $\Delta\varphi$; with independent clocks (P3) the accumulation $E_{\text{br}}\delta t_a/\hbar$ and $E_{\text{br}}\delta t_b/\hbar$ enters the difference. For internal branches $|0\rangle, |1\rangle$ with a common carrier $|C\rangle$ we take $E_{\text{br}} = \Delta E$ and $\omega_{\text{eff}} = \Delta E/\hbar$; since here $E_a = 0$, $E_b = \Delta E$, dephasing is twice as weak as in the symmetric case: $\chi = \omega_{\text{eff}}^2 \delta t^2/2$.

Since $\delta t_a, \delta t_b$ are Gaussian (P3), the relative phase $\Delta\varphi$ is also Gaussian and $\langle e^{i\Delta\varphi} \rangle = \exp[-\text{Var}(\Delta\varphi)/2]$; the factor 2 from the two

branches and the factor 1/2 from Gaussian averaging cancel. In the which-path case:

$$\frac{V}{V_0} = e^{-\chi}, \quad \chi = \omega_{\text{eff}}^2 \delta t^2 = \left(\frac{mc^2}{\hbar} \right)^2 (\xi t_P^2 t)^{2/3}. \quad (6)$$

The general expression for an arbitrary branch pair is $\chi = (E_a^2 + E_b^2) \delta t^2 / (2\hbar^2)$; for $E_a = E_b = mc^2$ it reduces to (6). The scaling form $\chi \propto m^2 t^{2/3}$ in a centre-of-mass interferometer follows from the postulates and constitutes the central testable prediction; the overall coefficient carries an $\mathcal{O}(1)$ uncertainty (through $\xi^{2/3}$, the interpretation of δt , and the choice of the effective t ; see § 4). The mass-quadratic dependence matches the general expectation for conjectural space-time fluctuations in the matter-wave interferometry literature [13].

Relation to clock interference and the Compton phase. The conceptual framework of “proper time along each branch” is already used in quantum-clock interference. In Zych et al., an internal state acts as a clock: a deterministic proper-time difference along the two paths records which-path information in the internal degree of freedom and reduces visibility [14]. Piovski et al. developed this mechanism for a composite object in a gravitational field: an externally imposed potential difference unitarily entangles the centre of mass with the internal-energy distribution, and dephasing appears after tracing out the internal degrees of freedom [15]. In contrast, the present model requires neither an external potential nor an internal clock, introduces the stochastic Planckian scale P2 and branch clock independence (P3), and therefore predicts irreversible ensemble dephasing even for an object in its internal ground state.

Using $\omega_C = mc^2/\hbar$ requires a separate caveat. In the “Compton clock” debate, atom interferometry has been interpreted as providing access to the mass frequency [16], whereas Wolf, Blanchet, and co-workers stressed that in standard quantum mechanics and metric theories the Compton phase cancels in a closed interferometer and does not by itself constitute an observable clock rate [17]. The present model does not settle that debate within standard theory; it adds the more radical premise that independent $\delta t_a, \delta t_b$ turn the common mc^2 term into a random *relative* phase contribution. The energy zero is not arbitrary

here: P1 selects physical rest mass relative to the vacuum, while adding an arbitrary constant to an effective Hamiltonian must not change E_{br} . This is an independent, testable postulate; if the rest phase does not become operational under independent clocks, the central prediction $\chi \propto m^2$ does not arise.

Planck mass boundary. The decoherence time t_{dec} is defined by the condition $\chi = 1$. From (6):

$$t_{\text{dec}} = \frac{t_C^3}{\xi t_P^2}, \quad t_C = \frac{\hbar}{mc^2}. \quad (7)$$

Setting $t_{\text{dec}} = t_C$, we obtain $t_C = \sqrt{\xi} t_P$, i.e. the characteristic scale

$$m_* = \frac{m_P}{\sqrt{\xi}}, \quad m_P = \sqrt{\hbar c/G} \approx 2.2 \times 10^{-8} \text{ kg}. \quad (8)$$

For $\xi = \mathcal{O}(1)$ this is of the order of the Planck mass. For $m \gg m_*$ the decoherence time $t_{\text{dec}} \ll t_C$, and coherence between branches is lost over an interval of microscopic duration—in principle indistinguishable and unmeasurable, since it lies below the uncertainty δt from (2). Schrödinger’s cat ($m \sim 1 \text{ kg} \gg m_P$) is the limiting example: the “alive/dead” superposition is amplitudinally recordable, but interference between branches vanishes over t_{dec} , inaccessible to any clock; in the ensemble reading what remains is a statistical mixture with weights $|c_i|^2$. The mechanism remains pure dephasing: populations are not redistributed, and there is none of the spontaneous emission predicted by Diósi–Penrose models [18]. Ontologically, however, once $\chi \gg 1$ the loss of coherence renders the alternatives operationally classical: one branch is realized as a fact (with weight $|c_i|^2$), while the definite value becomes known only upon registration. In this restricted sense the macroscopic limit of the model plays the role of an objective collapse, differing from dynamical collapse models in the channel (rest-phase dephasing rather than a localizing jump) and in the absence of their characteristic side signals.

4 Confrontation with experiment

4.1 Matter-wave interferometry: the primary probe

By construction (P1)/(P3) the noise attaches to discrete branches, not to spatial points; the mechanism is therefore pure rest-phase dephasing: there is no position-dependent stochastic potential, hence no force, no momentum diffusion, and no spontaneous emission. We stress that this property is secured by the branching postulate (P1), not automatically: noise attached to positions would induce a force. Consequently, the model is *transparent* to the non-interferometric bounds that constrain Diósi–Penrose: the underground search for spontaneous radiation [18], the Majorana Demonstrator limit [19], and X-ray tests [20] do not apply to it. Nor does the mechanism induce any detectable instability in the rate of atomic clocks: the effect exists only *between* branches of a superposition, whereas single (non-superposed) clocks accumulate no rest phase differentially, so their rate is unaffected. Matter-wave interferometry of the centre of mass is therefore the most direct laboratory probe of the mechanism with $E_{\text{br}} = mc^2$.

For the same reason, purely optical interferometers provide no direct constraint: for a massless carrier $d\tau = 0$, so the photonic phase acquires no BDPT contribution. LIGO-like detectors constrain models of external metric fluctuations, while the Fermilab Holometer searched for correlated spatial shear noise and excluded a specific class of such holographic models [21]; the present model predicts neither signal. This does not make the entire optical apparatus immune: massive mirrors or beam splitters would be subject to the mechanism if they themselves occupied distinguishable centre-of-mass superpositions. They do not do so in the usual classical optical regime, and existing optical null results therefore place no direct bound on ξ . The absence of photonic and external metric noise distinguishes the model from spatially attached implementations of Planckian uncertainty.

4.2 Sodium nanoparticles: the Vienna group (2026)

Setup parameters. We confront Equation (6) with the sodium-nanoparticle interferometer of the Vienna group [1, 22]:

- clusters of 5000–10000 Na atoms with a mass-distribution centre $m \approx 172$ kDa ($m \approx 2.86 \times 10^{-22}$ kg);
- velocity $v \approx 160$ m/s;
- three gratings of period $d = 133$ nm spaced by $L = 0.983$ m;
- a measured fringe visibility $V = 0.10 \pm 0.01$ (second run 0.08 ± 0.01) at 172 kDa, described by the quantum model with a single global factor 0.78 for conventional contrast losses (alignment, gravitational and rotational phase averaging, vibrations, thermal and collisional decoherence);
- a record macroscopicity $\mu = 15.45$ [1, 23].

The relevant superposition time is the Talbot flight of one section $t = L/v \approx 6.1$ ms (justification of the choice—in § 4.4).

Under the sharpened P1 this is not a separate counting rule: the two resolved branches of a rigid cluster are translations of all its atoms. When their one-atom spatial supports have negligible overlap, $\mathcal{D}_{\text{Na}} = N$ and Equation (1) gives $E_{\text{br}} = Nm_{\text{at}}c^2 = Mc^2$, whence $\chi \propto M^2$. If the overlap is incomplete, M in the formulas below must be replaced by $m_{\text{at}}\mathcal{D}_{\text{Na}}$; the assumption of full branch distinguishability is thus explicit and can be checked against the packet geometry.

4.3 Composite superpositions: flux qubits and atomic NOON states

The unqualified phrase “energy of the split subsystem” is particularly hazardous for a superconducting flux qubit. Naively counting all 10^6 – 10^{10} electrons carrying superconductivity would give, at the upper end, $E_{\text{br}} \simeq 8.2 \times 10^{-4}$ J and, for $t = 100$ μ s and $\xi = 1$, $\chi \simeq 26.5$; the observed coherence of current states would then exclude the model. A microscopic analysis of the BCS states, however, shows that the near-unit overlap of the one-electron states in the two current branches leaves only a small fraction of occupations changed. An estimate equivalent to the one-particle distinguishability in (1) gives $\mathcal{D}_e \simeq$

42–5750 for the devices considered, rather than the total number of condensate electrons [24, 25]. At $t = 100 \mu\text{s}$ this gives $\chi \lesssim 9 \times 10^{-12}$: existing flux qubits are safe for the model. The difference between naive and microscopic counts is not an $\mathcal{O}(1)$ uncertainty; this is why the distinguishability measure is part of P1.

The same rule separates two often conflated regimes of atom interferometry. A product state

$$\left[(|L\rangle + |R\rangle)/\sqrt{2} \right]^{\otimes N} \quad (9)$$

contains independent one-atom superpositions; observed first-order coherence compares components with $\mathcal{D} = 1$ and acquires the atomic, not collective, energy mc^2 . In contrast, a NOON state

$$(|N, 0\rangle + |0, N\rangle)/\sqrt{2} \quad (10)$$

has two collective branches with $\mathcal{D} = N$ and $E_{\text{br}} = Nmc^2$. For $N = 10^5$ ^{87}Rb atoms and $t = 10$ ms, Equation (6) predicts $\chi \simeq 1.4 \times 10^3$ at $\xi = 1$, i.e. practically complete suppression of interference for a massive motional NOON state. This additional falsifiable prediction does not yet conflict with experiment: demonstrated atomic NOON states remain few-particle; for example, Ref. [26] realized a second-order state.

4.4 Numerical estimate and constraint on ξ

We first fix the effective superposition time. The postulates do not uniquely define the moment at which independent branch clocks “switch on”: delocalization of the wave function grows already after the first grating (full flight $2L/v \approx 12.3$ ms), but distinguishable diffraction orders—branches in the strict sense of (P1)—form only at the second grating, leaving one section until recombination at the third, $t = L/v \approx 6.1$ ms. For an upper bound we take the *minimal* interpretation $t = L/v$: the smaller t , the smaller the predicted dephasing and the *weaker* (more conservative) the resulting bound on ξ ; any broader interpretation of the superposition time would only tighten the limit. With $\omega_C = mc^2/\hbar = 2.43 \times 10^{29} \text{ s}^{-1}$ (at $m = 172 \text{ kDa}$) and $t_P^2 = 2.91 \times 10^{-87} \text{ s}^2$, Equation (6) gives

$$\chi \approx 0.41 \xi^{2/3} \quad (t = L/v \approx 6.1 \text{ ms}). \quad (11)$$

At $\xi = 1$ this is $\chi \approx 0.41$, i.e. a visibility deficit of $\approx 34\%$.

We use the published raw data and analysis code [27]. In the bin $P_2 = 15.252 \text{ mW}$ three independent scans give

$$V_i = 0.0732 \pm 0.0081, \quad 0.0975 \pm 0.0070, \quad 0.0780 \pm 0.0074, \quad (12)$$

where the errors are 1σ from the covariance matrix of a nonlinear sinusoidal fit. For the displayed rounded values, the inverse-variance mean is $\bar{V} = 0.0840$ with a formal error of 0.0043. For these three scans $\chi_{\text{red}}^2 = 3.08$; following the experiment’s procedure for overdispersed bins, we inflate the error by $\sqrt{\chi_{\text{red}}^2}$ and obtain $\sigma_{\bar{V}} = 0.0076$.

The official quantum curve at this power gives $V_{\text{QM}} = 0.1108$ (without conventional losses). The one-sided lower bound on the observed visibility at 95% is

$$\bar{V} - 1.645 \sigma_{\bar{V}} = 0.0716. \quad (13)$$

Here and below, numerical combinations are evaluated from the unrounded intermediate values ($\bar{V} = 0.0840247$, $\sigma_{\bar{V}} = 0.0075552$) and only then rounded for display.

We construct the upper bound on ξ strictly. The observed visibility decomposes as $V_{\text{obs}} = R_{\text{ord}} V_{\text{QM}} e^{-\chi}$, where $R_{\text{ord}} \leq 1$ is the total conventional contrast-loss factor (alignment, vibrations, phase averaging, gas, radiation). Since R_{ord} cannot exceed unity by its physical meaning, regardless of its true value we have $e^{-\chi} \geq V_{\text{obs}}/V_{\text{QM}}$. Attributing *the entire* visibility deficit to our mechanism ($R_{\text{ord}} = 1$), we obtain

$$e^{-\chi} \geq \frac{0.0716}{0.1108} = 0.646, \quad \chi \leq 0.44, \quad (14)$$

and from $\chi = 0.41 \xi^{2/3}$ —a strict conservative bound

$$\boxed{\xi \lesssim 1.1 \quad (95\% \text{ CL, one-sided}).} \quad (15)$$

This bound does not depend on the experimental calibration: any real conventional losses ($R_{\text{ord}} < 1$) could only tighten it. The exclusion threshold corresponds to a visibility deficit $1 - 0.646 \approx 35\%$: values of ξ predicting a larger deficit are incompatible with the data at 95%.

If one trusts the published estimate of conventional losses $R_{\text{ord}} = 0.78$, the limit tightens: $e^{-\chi} \geq 0.0716/(0.78 \cdot 0.1108) = 0.828$, $\chi \leq 0.188$, whence $\xi < 0.31$. But this is a conditional result: at a single mass the product $R_{\text{ord}} e^{-\chi}$ is degenerate, and the data do not allow separating the new mechanism from underestimated conventional losses; the interpretation of δt , the choice of effective t , and the overall postulate coefficient also remain model systematics.

Summary for the data: at $\xi = 1$ the model predicts ($R_{\text{ord}} = 1$) visibility $0.1108 e^{-0.41} \approx 0.074$ against the observed 0.0840 ± 0.0076 —a deviation of only 1.4σ , so $\xi = \mathcal{O}(1)$ is *not excluded* by the data; the strict bound presses the coefficient to $\xi \lesssim 1.1$, the conditional one (with the 0.78 calibration) to 0.31. A statistically rigorous limit requires a joint likelihood analysis of the raw scans, the quantum-curve parameters, and all conventional contrast-loss channels.

4.5 The decisive near-term test

The already-measured high visibility $V = 0.66 \pm 0.09$ at 0.4–1 MDa [1] does *not* test the mechanism as distinguishable quantum coherence: it was taken in the regime $\lambda_{dB} \lesssim 3$ fm, $L \leq L_T$ ($L_T \sim d^2/\lambda_{dB}$ —the *Talbot length*, the scale of grating self-imaging in the near field of a Talbot–Lau interferometer [1, 13]), where the quantum and classical predictions coincide (geometric optics). A discriminating test requires the quantum regime at high mass. The same programme targets $m \sim 1$ MDa with $v \approx 25$ m/s [1]. Then even with the conservative $t = L/v \approx 39$ ms and $\omega_{\text{eff}} \approx 1.4 \times 10^{30} \text{ s}^{-1}$, Equation (6) at $\xi = 1$ gives

$$\chi \approx 47, \quad V/V_0 = e^{-\chi} \lesssim 10^{-20}. \quad (16)$$

Practically complete loss of interference is robust against $\mathcal{O}(1)$ uncertainty in the coefficient (even dividing χ by $\sim \pi^2$ leaves $\chi \sim 5$). The standard quantum model of the setup at the same parameters expects measurable fringes with ordinary contrast. Observing fringes at ~ 1 MDa and $v \approx 25$ m/s would falsify the model.

4.6 Distinguishing signature

A visibility deficit alone is not diagnostic, since thermal radiation, collisions with residual gas, and

velocity spread also reduce the contrast. The signature of the present mechanism is its scaling $\chi \propto m^2 t^{2/3}$ together with *immunity to shielding*: it cannot be reduced by improving the vacuum or cooling the particle. The experimental discriminator is whether a residual deficit persists after the suppression of all known environmental channels and whether it follows the m^2 law. Disappearance of fringes at ~ 1 MDa and $v \approx 25$ m/s with scaling $V \propto \exp[-(mc^2/\hbar)^2 (\xi t_P^2 t)^{2/3}]$ after excluding environmental losses would be evidence in favour of the model.

An additional discriminator is the dependence on the maximum spatial branch separation Δx . For momentum-transfer decoherence (collisions and photon emission or scattering), the localization exponent contains factors of the form $1 - \text{sinc}(q\Delta x/\hbar)$: it vanishes for unresolved paths, grows as the environment starts to resolve them, and saturates at the event rate once Δx exceeds the environment’s spatial resolution [28]. It would therefore be incorrect to say that *all* environmental channels grow without bound with Δx . A gravitational gradient usually produces a deterministic geometry-dependent phase; contrast is lost only after averaging over fluctuations of the gradient or trajectories.

The present model is likewise not strictly independent of Δx over the entire range. Under the sharpened P1,

$$\chi_{\text{BDPT}}(\Delta x) = \left[\frac{c^2 \sum_s m_s \mathcal{D}_s(\Delta x)}{\hbar} \right]^2 (\xi t_P^2 t)^{2/3}. \quad (17)$$

While the packets overlap, $\mathcal{D}_s(\Delta x)$ grows from zero. The specific signature emerges once the branches are resolved: as $\mathcal{D}_s \rightarrow N_s$, increasing Δx further does not change χ_{BDPT} , producing a plateau that depends only on m and t . This is the direct consequence of attaching the noise to distinguishable branches rather than to spatial points.

The test can be performed in one apparatus: at fixed m , v , superposition time, pressure, and temperature, vary the transverse momentum transfer (preferably the diffraction order; changing the grating period also requires compensating the change in Talbot length) and thereby scan Δx . The quantity to compare is not raw visibility but

the residual exponent

$$\chi_{\text{res}}(\Delta x) = -\ln \left[\frac{V_{\text{obs}}(\Delta x)}{V_{\text{QM}}(\Delta x)} \right] - \chi_{\text{env}}^{\text{calc}}(\Delta x), \quad (18)$$

where V_{QM} includes coherent dynamics and grating interactions, while $\chi_{\text{env}}^{\text{calc}}$ contains the calculated gas and thermal contributions. The model predicts a constant addition to χ_{res} on the distinguishability plateau, whereas unsaturated environmental channels retain a characteristic Δx dependence. A flat residual alone is not unique: saturated environmental decoherence and Δx -independent technical losses can mimic it. The Δx scan should therefore be combined with changes in vacuum or temperature and with a test of the $m^2 t^{2/3}$ scaling.

5 Limits on quantum computation

5.1 Margolus–Levitin: a heuristic comparison

The decoherence time of the model is the moment when $\chi = 1$ and the visibility falls to e^{-1} ; it is given by Equation (7).

The Margolus–Levitin theorem [29] limits the speed of orthogonal evolution by the energy *above the ground level* $E_{\text{av}} = \langle H \rangle - E_0$: $\tau_{\perp} \geq \pi \hbar / (2E_{\text{av}})$, whence $\nu_{\text{max}} = 2E_{\text{av}} / (\pi \hbar)$. This is a question *distinct* from branch decoherence: the Margolus–Levitin theorem concerns the unitary speed of state change, not the noise of independent clocks.

At fixed particle number, the term mc^2 is proportional to the identity and does not generate orthogonal centre-of-mass evolution; it cancels from $E_{\text{av}} = \langle H \rangle - E_0$. Thus the identification $E_{\text{av}} = E_{\text{br}} = mc^2$ follows neither from the theorem nor from the model’s postulates. Only as a formal dimensional illustration, under the additional and nonstandard assumption of that identification, one may write the number of “ticks” over the window t_{dec} as

$$N_{\text{max}} = \nu_{\text{max}} t_{\text{dec}} = \frac{2\omega_{\text{eff}}}{\pi} \cdot \frac{t_C^3}{\xi t_P^2} = \frac{2}{\pi \xi} \left(\frac{t_C}{t_P} \right)^2. \quad (19)$$

Since $t_C/t_P = m_P/m$, for which-path

$$N_{\text{max}}^{\text{formal}} = \frac{2}{\pi \xi} \left(\frac{m_P}{m} \right)^2. \quad (20)$$

At $m = m_* = m_P/\sqrt{\xi}$ this formal quantity is $2/\pi \approx 0.64$. It is not a computational-power bound derived from the model. For internal branches, where the transition energy actually generates evolution, the theorem should instead be applied directly using the physical E_{av} , without importing the formula based on mc^2 .

5.2 Formal coincidence with holographic scaling

The same additional identification gives the formal rate $N = (2/\pi) t/t_C$. Combining it with the coherence condition $t_{\text{dec}} \geq t$, i.e. $t_C \geq (\xi t_P^2 t)^{1/3}$, and taking equality at the boundary gives

$$N_{\text{formal}}(t) = \frac{2}{\pi \xi^{1/3}} \left(\frac{t}{t_P} \right)^{2/3}. \quad (21)$$

Over one second this is $\approx 4.5 \times 10^{28}$ at $\xi = 1$. The exponent $2/3$ coincides with the scaling $N \propto (t/t_P)^{2/3}$ obtained by Ng for a “space-time-foam computer” and a black-hole quantum computer [11], because both estimates use the cubic law $\delta t^3 \sim t_P^2 t$. This is a meaningful structural analogy, but not an independent derivation of an operation-count bound: without additional physical justification for $E_{\text{av}} = mc^2$, matching exponents does not turn N_{formal} into a computational limit of the model.

5.3 Phase noise and the qubit register

Postulates (P1)–(P3) act on any superposition, including a computational one. For strictly internal levels $|0\rangle, |1\rangle$ with a common one-particle spatial-motional density matrix, P1 gives $E_{\text{br}} = \hbar \omega_q = \Delta E$; the mass of the chip and resonator enters the common $|C\rangle$ and does not enter ω_{eff} . This statement cannot automatically be transferred to a superposition of opposite persistent currents, for which Equation (1) applies as discussed in § 4.3. Superconducting resonators storing a single photon for ~ 34 ms with $\gtrsim 10^{23}$ electrons in their walls [30] illustrate the internal

case: with $E_{\text{br}} = \Delta E$ of the photon, not the rest energy of the common walls, long-lived coherence is consistent with the model. The random relative phase between a pair of internal branches over time t :

$$\sigma(t) = \frac{\sqrt{E_a^2 + E_b^2}}{\hbar} \delta t = \omega_q (\xi t_P^2 t)^{1/3} \quad (E_a = 0, E_b = \hbar \omega_q) \quad (22)$$

For a superconducting qubit ($\omega_q \approx 2\pi \times 5$ GHz) and $t = 1$ s: $\delta t \approx 1.4 \times 10^{-29}$ s, $\sigma \approx 4.5 \times 10^{-19}$ rad (the corresponding dephasing parameter $\chi = \sigma^2/2$). Crucially, the variance (22) is *pairwise* and independent of the number of branches: interference involves only differences of independent δt_a , and the extremal phase spread over 2^n branches grows only as $\sigma \sqrt{2 \ln 2^n} \propto \sigma \sqrt{n}$. For a branch pair differing in the state of d qubits, $\chi \propto d^2$; in the worst case, when all n qubits differ, the formal limit $n \omega_q \delta t \lesssim 1$ gives $n \lesssim 10^{18}$ —far beyond practical horizons.

5.4 Phase distinguishability floor: exact and approximate QFT

The quantity σ sets a *floor* for algorithmic phase rotations: a gate at angle $\theta \lesssim \sigma$ is indistinguishable from the model’s intrinsic noise and cannot carry information. An exact quantum Fourier transform on n qubits contains controlled rotations down to $2\pi/2^n$, and already at $n \gtrsim 64$ these angles fall below the floor $\sim 10^{-19}$ rad: exact QFT on large registers is forbidden by the model. However, exact QFT is not used in implementations either: by Coppersmith [31] all rotations with $k > \mathcal{O}(\log(n/\varepsilon))$ can be *omitted from the circuit* entirely, losing only ε success probability; the minimum remaining angle $\sim 2\pi\varepsilon/n$ is polynomially, not exponentially, small. For Shor’s algorithm [32] at $n = 10^4$, $\varepsilon = 10^{-2}$ this is $\sim 10^{-5}$ rad—thirteen orders of magnitude above the floor (22). Phase-estimation algorithms that read the answer from the position of an interference peak are therefore not limited by the mechanism: it forbids precisely those exponentially small rotations that are excluded from circuits anyway.

5.5 Grover’s algorithm limit

The case of amplitude amplification [33] is contrasting. Here there is no QFT and nothing to truncate: each iteration rotates the state vector

by a fixed angle $2\theta \approx 2/\sqrt{N}$, set by the geometry of the problem, and all $\sim (\pi/4)\sqrt{N}$ iterations form one continuous coherent accumulation without intermediate readouts. The small rotation per step is irremovable: it *is* the algorithm. Comparability with the floor (22) arises when $2/\sqrt{N} \sim \sigma$, i.e. at $N \sim (2/\sigma)^2 \sim 2 \times 10^{37} \approx 2^{124}$: there the signal step is $2^{-61} \approx 4 \times 10^{-19}$ rad. At smaller N (e.g. $N = 2^{80}$, step $\sim 10^{-12}$ rad) the rotation is orders of magnitude above the floor and the mechanism is imperceptible on a single iteration; at larger N the step falls below the floor, and dephasing accumulated over $\sim \sqrt{N}$ iterations becomes critical—although in practice the computation time kills it first. This is consistent with the known structural fragility of the algorithm: with constant noise speed on the oracle, Grover’s quadratic speedup is destroyed completely [34].

6 Conclusion

Tying the Károlyházy fluctuation to proper time and postulating independent branch clocks yields a decoherence law (for which-path, $\xi = \mathcal{O}(1)$)

$$V/V_0 = \exp \left[- \left(\frac{mc^2}{\hbar} \right)^2 (\xi t_P^2 t)^{2/3} \right], \quad (23)$$

where interference involves the *difference* of branch phase contributions and mc^2 is the energy of the split carrier. The scaling $m^2 t^{2/3}$ is immune to electromagnetic shielding; sodium interferometry [1] gives a strict one-sided 95% bound $\xi \lesssim 1.1$ (the entire contrast deficit attributed to the mechanism; conditionally, with the conventional-loss normalization 0.78, $\xi \lesssim 0.31$). For $\xi = \mathcal{O}(1)$ the characteristic scale $m_* = m_P/\sqrt{\xi}$ is of the order of the Planck mass; for $m \gg m_*$ decoherence occurs over $t_{\text{dec}} \ll t_C$, below the Károlyházy distinguishability limit for proper time. The model is falsifiable by a test at ~ 1 MDa and $v \approx 25$ m/s: either fringes of the standard quantum model, or suppression with the $m^2 t^{2/3}$ scaling.

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References

- [1] Pedalino, S., Ramírez-Galindo, B.E., Ferstl, R., Hornberger, K., Arndt, M., Gerlich, S.: Probing quantum mechanics with nanoparticle matter-wave interferometry. *Nature* **649**, 866–870 (2026) <https://doi.org/10.1038/s41586-025-09917-9>
- [2] Ballentine, L.E.: The statistical interpretation of quantum mechanics. *Reviews of Modern Physics* **42**(4), 358–381 (1970) <https://doi.org/10.1103/RevModPhys.42.358>
- [3] Home, D., Whitaker, M.A.B.: Ensemble interpretations of quantum mechanics. A modern perspective. *Physics Reports* **210**(4), 223–317 (1992) [https://doi.org/10.1016/0370-1573\(92\)90088-H](https://doi.org/10.1016/0370-1573(92)90088-H)
- [4] Page, D.N., Wootters, W.K.: Evolution without evolution: Dynamics described by stationary observables. *Physical Review D* **27**, 2885–2892 (1983) <https://doi.org/10.1103/PhysRevD.27.2885>
- [5] Penrose, R.: On gravity’s role in quantum state reduction. *General Relativity and Gravitation* **28**(5), 581–600 (1996) <https://doi.org/10.1007/BF02105068>
- [6] Korsbakken, J.I., Whaley, K.B., DuBois, J., Cirac, J.I.: Measurement-based measure of the size of macroscopic quantum superpositions. *Physical Review A* **75**, 042106 (2007) <https://doi.org/10.1103/PhysRevA.75.042106>
- [7] Károlyházy, F.: Gravitation and quantum mechanics of macroscopic objects. *Il Nuovo Cimento A* **42**, 390–402 (1966) <https://doi.org/10.1007/BF02717926>
- [8] Salecker, H., Wigner, E.P.: Quantum limitations of the measurement of space-time distances. *Physical Review* **109**, 571–577 (1958) <https://doi.org/10.1103/PhysRev.109.571>
- [9] Ng, Y.J., Dam, H.: Limitation to quantum measurements of space-time distances. *Annals of the New York Academy of Sciences* **755**, 579–583 (1995) <https://doi.org/10.1111/j.1749-6632.1995.tb39004.x>
- [10] Ng, Y.J., Dam, H.: Limitation to quantum measurements of space-time distances. *Modern Physics Letters A* **9**, 335–340 (1994) <https://doi.org/10.1142/S0217732394000330>
- [11] Ng, Y.J.: Spacetime foam, holographic principle, and black hole quantum computers. *International Journal of Modern Physics A* **20**, 1109–1122 (2005) <https://doi.org/10.1142/S0217751X05022858> [arXiv:gr-qc/0403057](https://arxiv.org/abs/gr-qc/0403057)
- [12] Mandelbrot, B.B., Van Ness, J.W.: Fractional Brownian motions, fractional noises and applications. *SIAM Review* **10**(4), 422–437 (1968) <https://doi.org/10.1137/1010093>
- [13] Kiałka, F., Fein, Y.Y., Pedalino, S., Gerlich, S., Arndt, M.: A roadmap for universal high-mass matter-wave interferometry. *AVS Quantum Science* **4**, 020502 (2022) <https://doi.org/10.1116/5.0088913>
- [14] Zych, M., Costa, F., Piovski, I., Brukner, Č.: Quantum interferometric visibility as a witness of general relativistic proper time. *Nature Communications* **2**, 505 (2011) <https://doi.org/10.1038/ncomms1498> [arXiv:1105.4531](https://arxiv.org/abs/1105.4531)
- [15] Piovski, I., Zych, M., Costa, F., Brukner, Č.: Universal decoherence due to gravitational time dilation. *Nature Physics* **11**, 668–672 (2015) <https://doi.org/10.1038/nphys3366> [arXiv:1311.1095](https://arxiv.org/abs/1311.1095)
- [16] Lan, S.-Y., Kuan, P.-C., Estey, B., English, D., Brown, J.M., Hohensee, M.A., Müller, H.: A clock directly linking time to a particle’s mass. *Science* **339**(6119), 554–557 (2013)

<https://doi.org/10.1126/science.1230767>

- [17] Wolf, P., Blanchet, L., Bordé, C.J., Reynaud, S., Salomon, C., Cohen-Tannoudji, C.: Does an atom interferometer test the gravitational redshift at the Compton frequency? *Classical and Quantum Gravity* **28**, 145017 (2011) <https://doi.org/10.1088/0264-9381/28/14/145017> arXiv:1012.1194
- [18] Donadi, S., *et al.*: Underground test of gravity-related wave-function collapse. *Nature Physics* **17**, 74–78 (2021) <https://doi.org/10.1038/s41567-020-1008-4>
- [19] Arnquist, I.J., *et al.*: Search for spontaneous radiation from wave-function collapse in the Majorana Demonstrator. *Physical Review Letters* **129**, 080401 (2022) <https://doi.org/10.1103/PhysRevLett.129.080401>
- [20] Piscicchia, K., *et al.*: X-ray emission from atomic systems can distinguish between prevailing dynamical wave-function collapse models. *Physical Review Letters* **132**, 250203 (2024) <https://doi.org/10.1103/PhysRevLett.132.250203>
- [21] Chou, A.S., *et al.*: Interferometric constraints on quantum geometrical shear noise correlations. *Classical and Quantum Gravity* **34**, 165005 (2017) <https://doi.org/10.1088/1361-6382/aa7bd3> arXiv:1703.08503
- [22] Pedalino, S., Ramírez-Galindo, B.E., Ferstl, R., Hornberger, K., Arndt, M., Gerlich, S.: Probing quantum mechanics using nanoparticle Schrödinger cats (2025)
- [23] Nimmrichter, S., Hornberger, K.: Macroscopicity of mechanical quantum superposition states. *Physical Review Letters* **110**, 160403 (2013) <https://doi.org/10.1103/PhysRevLett.110.160403>
- [24] Korsbakken, J.I., Wilhelm, F.K., Whaley, K.B.: Electronic structure of superposition states in flux qubits. *Physica Scripta* **T137**, 014022 (2009) <https://doi.org/10.1088/0031-8949/2009/T137/014022> arXiv:0910.3622
- [25] Korsbakken, J.I., Wilhelm, F.K., Whaley, K.B.: The size of macroscopic superposition states in flux qubits. *EPL* **89**, 30003 (2010) <https://doi.org/10.1209/0295-5075/89/30003> arXiv:1003.5294
- [26] Chen, Y.-A., Bao, X.-H., Yuan, Z.-S., Chen, S., Zhao, B., Pan, J.-W.: Heralded generation of an atomic NOON state. *Physical Review Letters* **104**, 043601 (2010) <https://doi.org/10.1103/PhysRevLett.104.043601>
- [27] Pedalino, S., Ramírez Galindo, B.E., Ferstl, R., Hornberger, K., Arndt, M., Gerlich, S.: Probing quantum mechanics with nanoparticle matter-wave interferometry - Dataset and code. Zenodo (2025). <https://doi.org/10.5281/zenodo.17502163>
- [28] Hornberger, K., Sipe, J.E., Arndt, M.: Theory of decoherence in a matter wave Talbot–Lau interferometer. *Physical Review A* **70**, 053608 (2004) <https://doi.org/10.1103/PhysRevA.70.053608> arXiv:quant-ph/0407245
- [29] Margolus, N., Levitin, L.B.: The maximum speed of dynamical evolution. *Physica D* **120**, 188–195 (1998) [https://doi.org/10.1016/S0167-2789\(98\)00054-2](https://doi.org/10.1016/S0167-2789(98)00054-2) arXiv:quant-ph/9710043
- [30] Milul, O., Guttel, B., Goldblatt, U., Hazanov, S., Joshi, L.M., Chausovsky, D., Kahn, N., Çiftyürek, E., Lafont, F., Rosenblum, S.: Superconducting cavity qubit with tens of milliseconds single-photon coherence time. *PRX Quantum* **4**, 030336 (2023) <https://doi.org/10.1103/PRXQuantum.4.030336> arXiv:2302.06442
- [31] Coppersmith, D.: An approximate Fourier transform useful in quantum factoring. IBM Research Report RC 19642 (1994)
- [32] Shor, P.W.: Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM Journal on Computing* **26**(5), 1484–1509 (1997) <https://doi.org/10.1137/S0097539795293172>
- [33] Grover, L.K.: A fast quantum mechanical

algorithm for database search. In: Proceedings of the 28th Annual ACM Symposium on Theory of Computing (STOC), pp. 212–219 (1996). <https://doi.org/10.1145/237814.237866>

- [34] Regev, O., Schiff, L.: Impossibility of a quantum speed-up with a faulty oracle. In: Automata, Languages and Programming (ICALP 2008), Lecture Notes in Computer Science, vol. 5125, pp. 773–781 (2008). https://doi.org/10.1007/978-3-540-70575-8_63